

4-Month Rust Learning Roadmap

Month 1: Rust Basics + Tooling

Goal: Get comfortable with the Rust language and ecosystem

Topics:

- Rust installation (rustup, cargo, rustc)
- Basic syntax: variables, functions, control flow
- Ownership, Borrowing, and Lifetimes (Rust's core concept)
- Data types, Structs, Enums
- Pattern Matching (match, if let, while let)
- Error handling (Result, Option, unwrap, ?)
- Collections: Vec, HashMap, String

Resources:

- The Rust Book: <https://doc.rust-lang.org/book/>
- Rustlings Exercises: <https://github.com/rust-lang/rustlings>

Projects:

- CLI calculator
- To-Do app (CLI)

Month 2: Intermediate Concepts + Ownership Mastery

Goal: Deepen understanding of memory safety, traits, and project structure

Topics:

- Traits and Generics
- Lifetimes in detail
- Modules and project architecture
- Error handling (custom errors, thiserror, anyhow)
- Closures and iterators
- Smart pointers: Box, Rc, Arc, RefCell

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- Multithreading and async basics (tokio)

Resources:

- Rust by Example: <https://doc.rust-lang.org/rust-by-example/>
- Zero to Production in Rust: <https://www.zero2prod.com/>

Projects:

- File organizer CLI
- Multi-threaded file downloader
- Blog backend using actix-web or axum

Month 3: Web Development + Async Rust

Goal: Build web apps and APIs with Rust

Topics:

- tokio and async/await
- Web frameworks: Actix Web or Axum
- Working with databases: SQLx, Diesel, or SeaORM
- Auth (JWT, sessions)
- Caching with Redis
- Environment config (dotenv, config)

Resources:

- Axum Guide: <https://docs.rs/axum/latest/axum/>
- Tokio async runtime: <https://tokio.rs/>

Projects:

- REST API (Auth, CRUD, DB connection)
- Hacker News clone backend

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Month 4: Advanced Rust + Real-World Projects

Goal: Work on portfolio projects + Prepare for jobs

Topics:

- Unit and integration testing
- Logging (tracing, log)
- Code formatting and linting (rustfmt, clippy)
- FFI (optional): Rust + C interoperability
- WebAssembly (optional)
- Contributing to open-source Rust projects

Projects:

- Fullstack app using Rust backend + React/Next.js
- Chat app with WebSockets
- Publish a CLI tool on GitHub

Job Prep:

- Rust interview questions
- Solve problems on LeetCode with Rust
- Write a blog about your Rust journey

Bonus Tips

- Join Rust Discord: <https://discord.gg/rust-lang>
- Follow <https://users.rust-lang.org/>
- Contribute to open-source projects: ripgrep, tokei